

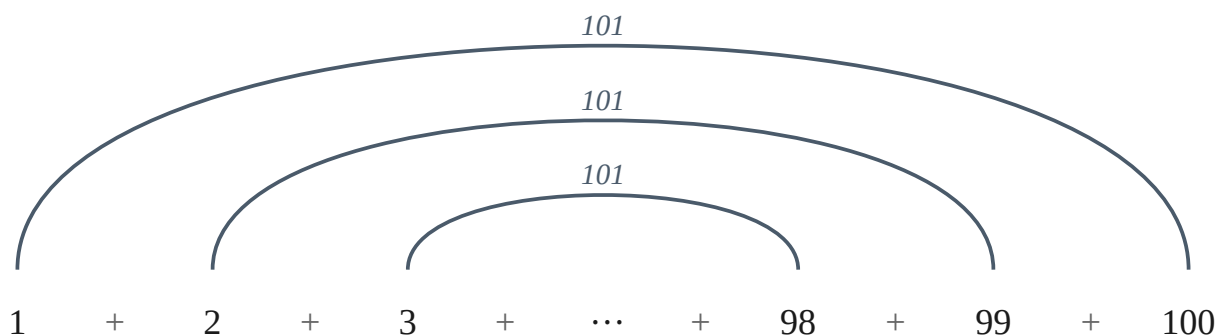
Technique Packet No. 1

Pairing — the trick a nine-year-old used to beat his teacher.

Name: _____ Date: _____

Almost 250 years ago in Germany, a schoolteacher wanted a few minutes of peace. So he handed his class what sounded like a punishment: add up every whole number from 1 to 100. Heads went down and chalk started scratching — 1, then $1 + 2 = 3$, then $3 + 3 = 6$, then $6 + 4 = 10$ — a long, slow march toward 100.

But one boy just sat and thought. A few seconds later he walked up and set his slate on the desk. On it was a single number: **5050**. The teacher was sure he had cheated — he hadn't. The boy was nine-year-old **Carl Gauss**, and he had seen something the marchers missed: pair the first number with the last, and $1 + 100 = 101$. Pair the next two in from the ends, and $2 + 99 = 101$. And $3 + 98 = 101$. Every pair makes the same total, 101 — and there are 50 of them, so the whole sum is just $50 \times 101 = 5050$. Gauss grew up to become one of the greatest mathematicians who ever lived, and this little move still carries his name.



Every pair from the outside in makes 101 — and there are 50 pairs, so $50 \times 101 = 5050$.

THE TECHNIQUE — Pairing.

To add a long, evenly spaced run of numbers:

1. Pair the first number with the last, the second with the second-to-last, and so on.
2. Check: every pair adds to the **same total**.
3. Count the pairs and multiply.
4. Odd count of numbers? The middle one has no partner — add it at the end.

When to reach for it: whenever a problem asks you to add a long, evenly spaced run of numbers — don't march, pair.

WORKED EXAMPLE 1**Find $1 + 2 + 3 + \dots + 20$.**

1. Pair the ends: $1 + 20 = 21$, $2 + 19 = 21$, $3 + 18 = 21$ — every pair makes 21.
2. The 20 numbers form 10 pairs.
3. So the sum is $10 \times 21 = \mathbf{210}$.

WORKED EXAMPLE 2 — when the count is odd**Find $1 + 2 + 3 + \dots + 9$.**

1. Pair the ends: $1 + 9 = 10$, $2 + 8 = 10$, $3 + 7 = 10$, $4 + 6 = 10$.
2. Nine numbers make 4 pairs — and the 5 in the middle has no partner.
3. The pairs give $4 \times 10 = 40$, and the leftover 5 brings it to 45.
4. So $1 + 2 + \dots + 9 = \mathbf{45}$.

Now you

1. Find $1 + 2 + 3 + \dots + 40$.

2. Find $5 + 6 + 7 + \dots + 14$.

3. Find $2 + 4 + 6 + \dots + 20$.

4. Find $1 + 2 + 3 + \dots + 99$.

5. A staircase is made of blocks. The first step is 1 block tall, the second is 2 blocks tall, the third is 3 blocks, and so on up to the 12th step. How many blocks does the whole staircase use?

6. Mia adds $1 + 2 + 3 + \dots$ and stops the moment her total reaches exactly 45. What is the last number she adds?

Answer Key — for parents

What this packet teaches. One named technique — pairing — shown first (story + two worked examples), then applied six times. The early problems look like the examples; the later ones hide the technique inside a situation, which is what contests do. The worked examples are written one sentence per line on purpose — they double as models of written explanations.

Answers.

1. 1–40: 20 pairs, each adding to 41 $\rightarrow 20 \times 41 = \mathbf{820}$.
2. 5–14 is ten numbers: 5 pairs of 19 (5+14, 6+13, ...) $\rightarrow \mathbf{95}$. The run not starting at 1 changes nothing — pairing only needs even spacing.
3. 2, 4, ... 20 is ten numbers: 5 pairs of 22 $\rightarrow \mathbf{110}$.
4. 1–99 is ninety-nine numbers — odd count, like Worked Example 2: 49 pairs of 100 plus the middle 50 $\rightarrow \mathbf{4950}$.
5. The blocks are $1 + 2 + \dots + 12$: 6 pairs of 13 $\rightarrow \mathbf{78}$. Nobody says “add a run of numbers” — they have to see the staircase *is* one.
6. Backwards: 45 is exactly Worked Example 2’s total, so her last number is **9**. Reaching 45 by pairing guesses (try $1-10 = 55$, too big; $1-9 = 45$) is just as good.

What to watch for. The marching habit — adding $1 + 2 = 3$, $+ 3 = 6$, $+ 4 = 10$... — gets the right answer on 1 and 2 and then collapses on 4. If you see marching, don’t correct it; wait for problem 4 to make the argument for you. On problem 2, the common slip is counting eleven numbers from 5 to 14 (it’s ten — count on fingers once).

How to run it. Read the story and the technique box together, out loud — then walk through both worked examples before they touch problem 1. The problems they do solo, in one sitting or two. This is a learning packet, not a test: helping on a stuck problem is fine, but help by pointing back at the worked examples, not at the answer.